Fiscal Year:	FY 2007	**Task Last Updated:**	FY 03/10/2008
PI Name:	Santhanakrishnan, Sivakumar Ph.D.		
Project Title:	Oxygen and Fuel Diffusion Flame Studies in Microgravity Motivated by Spacecraft Oxygen Storage Fire Safety		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Gaseous diffusion flames		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	sskris@iupui.edu	**Fax:**	FY 317 274-9744
PI Organization Type:	UNIVERSITY	**Phone:**	317 274-9711
Organization Name:	Indiana University - Purdue University Indianapolis		
PI Address 1:	723 W. Michigan St.		
PI Address 2:	SL 260H		
PI Web Page:			
City:	Indianapolis	**State:**	IN
Zip Code:	46202	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
Start Date:	10/01/2003	**End Date:**	09/30/2007
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	2
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA GRC
Contact Monitor:	Stocker, Dennis P	**Contact Phone:**	216-433-2166
Contact Email:	dennis.p.stocker@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 9/30/2007, per PI (10/06)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Gore, Jay (Purdue University) Sunderland, Peter (University of Maryland, College Park)		
Grant/Contract No.:	NNC04AA08A		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Normal and inverse diffusion flames will be considered burning hydrocarbon fuel in oxidizers of 21-100% O2 in N2. Diagnostics will include color video, radiometry, and multi-wavelength soot pyrometry in normal gravity and microgravity. Tests in normal gravity also will be performed and will include measurements of soot extinction and spectral emissions. The measurements will be used to validate coupled soot chemistry and radiation computations. Task Objective: 1. Examine the effects of enhanced oxygen conditions and inverse burning on microgravity laminar gas jet diffusion flames. 2. Consider the implications of these burning conditions on spacecraft fire safety. 3. Perform computational		

	studies to supplement the experimental work. One goal central to NASA's mission is to improve spacecraft fire safety for space shuttles and the space station. The nearly catastrophic Mir fire started when a canister leaked heated oxygen into the cabin. This work is expected to reveal new insights into possible fire scenarios on spacecraft and may lead to improved ways to prevent, detect, and extinguish them. In addition, advances into fundamental combustion configurations are anticipated.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This study is of relevance to fire safety in terrestrial environments where oxygen enhanced combustion is used and in emergency rooms in hospitals etc. where oxygen storage and use is common.
Task Progress:	34 normal diffusion flames (NDFs) and 38 inverse diffusion flames (IDFs) were established in earth-gravity and their high resolution photographs were taken with a view to obtaining flame length information for verifying the applicability of the Roper flame length correlations for normal and inverse flames. The total and spectral radiation properties of 16 NDFs and 7 IDFs were measured. This is one of the first studies on the total and spectral radiation properties of oxygen enhanced inverse diffusion flames. The total radiation was measured using a radiation flux meter and the spectral radiative intensity was measured using a fast infrared array spectrometer.
Bibliography Type:	Description: (Last Updated: 03/10/2008)
Abstracts for Journals and Proceedings	Saini MK, Krishnan SS, Zheng Y, Gore JP. "Effect of Oxygen Enhancement on Radiation Characteristics of Normal and Inverse Diffusion Flames." Spring Technical Meeting of the Combustion Institute Western States Section, San Diego, CA, March 2007. Proceedings, Spring Technical Meeting of the Combustion Institute Western States Section, San Diego, CA, March 2007. , Mar-2007
Articles in Peer-reviewed Journals	Krishnan SS, Abshire JM, Sunderland PB, Yuan Z-G, Gore JP. "Analytical Predictions of Laminar Diffusion Flame Shapes in Microgravity and Earth Gravity." Accepted for publication, Combustion Theory and Modeling, February 2008. , Feb-2008
Articles in Peer-reviewed Journals	Bhatia P, Katta VR, Krishnan SS, Zheng Y, Sunderland PB, Gore JP. "Simulations of Normal and Inverse Laminar Diffusion Flames under Oxygen Enhancement and Gravity Variation." Combustion Theory and Modeling, submitted and under review, February 2008. , Feb-2008
Dissertations and Theses	Saini MK. "Spectral and Total Radiation Properties of Oxygen Enhanced Normal and Inverse Laminar Diffusion Flames." MS Thesis, Purdue University, West Lafayette, IN, May 2006. , May-2006
Dissertations and Theses	Bhatia P. "Normal and Inverse Laminar Jet-Diffusion Flames under Oxygen Enhancement and Gravity-Variation." PhD Thesis, Purdue University, West Lafayette, IN (under review by the Committee, February 2008). , Feb-2008